

$$\text{best}(i, j) = \max \left\{ \begin{array}{l} \text{best}(i, j-1) \\ \max_{i \leq k < j; x_k, x_j \text{ pairable}} \text{best}(i, k-1) + \text{best}(k+1, j-1) + 1 \end{array} \right.$$

1. One-nonterminal weighted CFG for Nussinov folding.

$$S \rightarrow S(S) \text{ 1 } \mid S \cdot \text{ 0 } \mid \epsilon \text{ 0} \quad \text{red=weight}$$

$$\text{CKY} \rightarrow \text{best}(i, j) = \max \left\{ \begin{array}{l} \text{best}(i+1, j-1) + 1 \\ \max_{i \leq k < j} \text{best}(i, k) + \text{best}(k+1, j) \end{array} \right.$$

2. One-nonterminal weighted CFG for CKY folding.

$$S \rightarrow (S) \text{ 1 } \mid SS \text{ 0 } \mid \cdot \text{ 0 } \mid \epsilon \text{ 0}$$

3. Two-nonterminal weighted CFG for Nussinov
P nonterminal for (i, j) pair.

$$\begin{aligned} S &\rightarrow S \cdot \text{ 0 } \mid SP \text{ 0 } \mid \epsilon \text{ 0 ?} \\ P &\rightarrow (S) \text{ 1} \end{aligned}$$

4. P_{ij} means a pair exists. Pair P_{ij} has weight $w_{ij} = e^{-E_{ij}/RT}$

$$\begin{aligned} S &\rightarrow S \cdot \text{ 1 } \mid SP \text{ 1 } \mid \epsilon \text{ 1} \\ P &\rightarrow (S) w_{ij} \end{aligned} \quad \begin{array}{l} \text{way to view} \\ \text{CFG} \end{array}$$

Inside: $Q_{ij}(S)$ as inside partition func for S_{ij} and
 $Q_{ij}(P)$ as inside partition func for P_{ij}

$$\text{Base Case} \rightarrow Q_{i,i-1}(S) = 1$$

unpaired case:

$$Q_{ij}(S) = Q_{i,j-1}(S)$$

Paired case:

$$S \rightarrow SP$$

Left part: $S_{i,k-1}$ Paired Part: $P_{k,j}$

$$\text{Combined} = Q_{i,k-1}(S) Q_{k,j}(P)$$

$$\text{Sum over all } k\text{'s: } \sum_{k=i}^{j-1} Q_{i,k-1}(S) Q_{k,j}(P)$$

finally

$$P \rightarrow (S)$$

$$Q_{ij}(P) = w_{ij} \otimes Q_{i+1,j-1}(S)$$

Putting it all together

$$Q_{ij}(S) = Q_{i,j-1}(S) + \sum_{k=i}^{j-1} Q_{i,k-1}(S) Q_{k+1,j-1}(S) w_{kj}$$

Outside:

Base Case: $\hat{Q}_{1,n}(S) = 1$

Unpaired case: $S \rightarrow S.$

$$\hat{Q}_{i,j-1}(S) += \hat{Q}_{i,j}(S)$$

Paired case:

$$S \rightarrow SP$$

Left side: $S_{i,n-1}$ Pair: $P_{k,j}$

Left side outside: $\hat{Q}_{i,n-1}(S) += \hat{Q}_{i,j}(S) Q_{k,j}(P)$

Pair: $P_{k,j}$

Pair outside: $\hat{Q}_{k,j}(P) += \hat{Q}_{i,j}(S) Q_{i,k-1}(S)$

finally

$$P \rightarrow (S)$$

which is $P_{k,j} \rightarrow S_{n+1,j-1}$

$$\hat{Q}_{k+1,j-1}(S) = \hat{Q}_{k,j}(P) w_{k,j}$$

Putting inside and outside to simplify P_{ij} (base pair probs)

for pair (i,j) outside weight = $\hat{Q}_{i,j}(P)$ inside weight = $Q_{i,j}(P)$
total = $\hat{Q}_{i,j}(P) Q_{i,j}(P)$

total partition = $Q_{1,n}(S)$ so

$$P_{ij} = \frac{\hat{Q}_{i,j}(P) Q_{i,j}(P)}{Q_{1,n}(S)}$$

